

Pricing trends in multi-function scopes

"We see that the price trend of multi-function oscilloscope is moving towards affordability. For example, the range of Tektronix MDO3000 series of mixed domain oscilloscopes starts at ₹ 248,000."

—**Naresh Narasimhan, country marketing manager, Tektronix India**

"Keeping in mind the value addition that our multi-function scopes provide, prices have been stable and competitive with an increase in improvements and capabilities of oscilloscopes."

—**Sanchit Bhatia, application engineer, Keysight Technologies India Pvt Ltd**

"Prices of multi-function oscilloscopes are falling as critical components like analogue-to-digital (A-to-D), digital-to-analogue (D-to-A) converters, memory and high-performance field-programmable gate arrays (FPGAs) have become affordable mainstream items. PC based oscilloscopes, that take advantage of host PC processing and graphics capabilities, are challenging traditional benchtop instruments that have fixed size and resolution displays, with the additional benefit of delivering professional waveform analysis functionality right on designers' desktops."

—**Trevor J. Smith, business development manager, Pico Technology**

Naresh Narasimhan, country marketing manager, Tektronix India, informs, "The six-in-one instrument includes oscilloscope functionality, digital channels (MSO), protocol analysis capability, an 8-digit hardware counter/totaliser, a WaveGen function/arbitrary waveform generator and a digital voltmeter."

He adds, "This multi-use instrument provides design engineers and educational research the functionality of six independent instruments without the significant cost of separate instruments. In addition to saving cost, MDO3000 saves space on engineers' benches, improves usability with the same basic controls for multiple instruments and ensures engineers always have access to all the instruments they need."

Keysight recently introduced a power analyser that incorporates oscilloscope capabilities in a single instrument, looking to ease designing and testing procedures for R&D engineers on electronic power-conversion systems. While testing devices such as battery systems, electronic ballasts, appliances, power supplies, inverters and converters, there could arise a situation where one would need to test and measure power consumption under dynamic conditions. IntegraVision power analyser looks to meet those requirements as well as eliminate the need for an oscilloscope to visualise

repetitive and single-shot events.

Upgradable to high bandwidths.

When engineers are creating high-speed digital designs or even components, they require oscilloscopes capable of validating, debugging and optimising their designs, running pre-compliance tests, identifying weak points and maximising design margins. In addition to touchscreen and graphical trigger capabilities, manufacturers are offering oscilloscopes with ultra-high bandwidths ranging up to 100GHz for greater insights for validation and debugging.

For advanced analysis and surpassing measurement accuracy, vendors like Keysight are providing oscilloscopes that range from 8GHz to 33GHz, thereby allowing circuit designers and engineers to achieve quicker progress from first silicon to eventual product release.

On the other hand, Tektronix introduced a 70GHz bandwidth real-time oscilloscope with its patented asynchronous time interleaving (ATI) technology, which maintains signal-to-noise ratio for higher fidelity. This helps engineers who are involved in making high-speed coherent optical systems or research.

In October 2014, Teledyne LeCroy showcased their LabMaster 10-100Zi oscilloscope by competitively pushing real-time bandwidth and sample rate boundaries to achieve 100GHz and

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